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(AN UGC AUTONOMOUS INSTITUTION)
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# DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING (DATA SCIENCE)

## AI revolutionizing E-commerce Governance for Fraud Detection

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# ABSTRACT

Artificial Intelligence (AI) is revolutionizing e-commerce regulation through more effective and accurate fraud detection and customer retention. Global e-commerce fraud loss totaled more than $48 billion in 2024, with account takeovers representing more than 33% of the losses. Moreover, 63% of online merchants reported an increase in fraud activity, with 70% of them still employing legacy manual detection techniques. Time-consuming manual review techniques are non-deterministic and are unable to process large volumes of transactions, resulting in late detection and inability to detect patterns of fraud. To address these limitations, we present a multi-stage AI-based method for fraud detection and customer retention from a real-world transactional dataset. Data preprocessing is first applied to handle missing values, normalize numerical features, and encode categorical features.

# INTRODUCTION

The rise of digital commerce has reshaped the global retail landscape, providing consumers with unprecedented convenience and accessibility. International e-commerce sales hit an estimated $6.3 trillion in 2024, and projections show this amount will surpass $7.5 trillion by 2026, marking the accelerating transition to online shopping. Along with this growth, however, the complexity and magnitude of online transactions have made e-commerce platforms ever more vulnerable to fraud. The Global Fraud Index reported e-commerce fraud attempts grew 18% year-over-year, with certain types of fraud such as account takeovers, card-not-present fraud, and synthetic identity fraud experiencing drastic growth.

# MOTIVATION

The rapid expansion of the e-commerce sector has been driven by technological innovation and changing consumer behavior, especially post-pandemic. Industry giants like Amazon, Flipkart, Alibaba, and Shopify have seen a tremendous rush of customers, processing millions of transactions daily. This exponential customer increase has not only boosted revenues but also made these platforms prime targets for fraudulent activity. With the ease of online shopping, digital payments, and global shipping, e-commerce businesses must now manage an immense volume of real-time data. This creates both an opportunity and a challenge—while companies can personalize user experiences and maximize customer satisfaction, they also face the daunting task of monitoring for fraudulent behavior on a massive scale.

# PROBLEM DEFINATION

In the evolving landscape of e-commerce, the reliance on manual approaches for fraud detection and customer segmentation has become increasingly problematic and inefficient. As e-commerce transaction volumes grow and fraud patterns become increasingly complex, traditional manual systems are proving inadequate. These methods rely on static rules and human judgment, lacking the speed, precision, and scalability needed for real-time detection. The result is delayed identification of fraud, allowing malicious activities to proceed undetected, causing financial losses and eroding customer trust.

# LITERATURE SURVEY

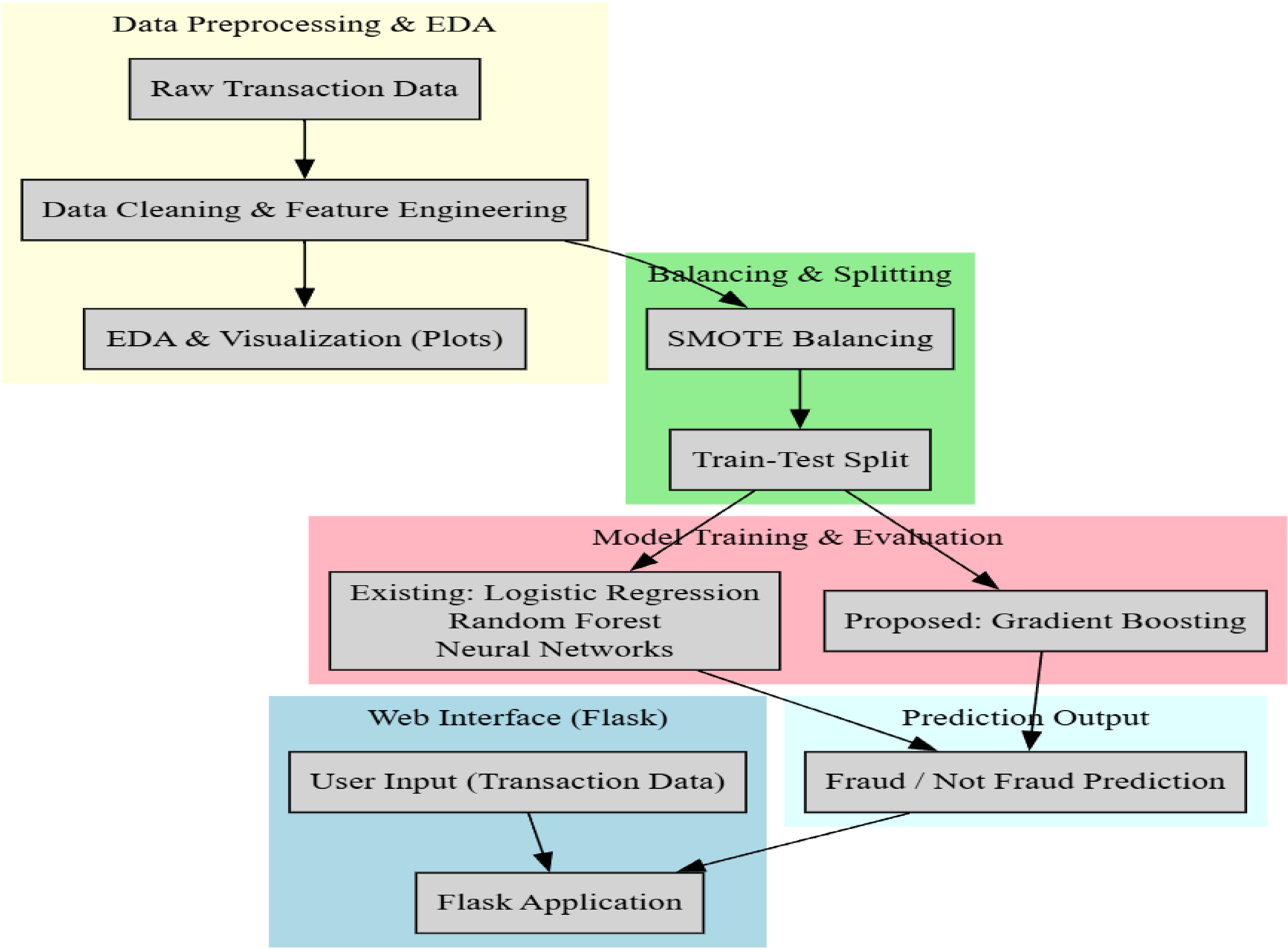
| AUTHOR | TITLE | YEAR | WORK DONE |
|---|---|---|---|
| Hasan et al. | "AI-driven fraud detection and mitigation in ecommerce transactions." | 2022 | proposed digital transactions pose a significant challenge to enterprises and customers for having secure business and financial transactions. |
| Khurana et al. | "Fraud detection in e-commerce payment systems: The role of predictive ai in real-time transaction security and risk management." | 2020 | analyzed huge datasets, found anomalies, and flagged possible fraud activities, thus enabling systems to make autonomous decisions during the process of payment. |
| Karunaratne et al. | "Machine Learning and Big Data Approaches to Enhancing Ecommerce Anomaly Detection and Proactive Defense Strategies in Cybersecurity." | 2023 | explored the synergistic application of machine learning and big data to enhance e-commerce cybersecurity. |

# RESEARCH GAPS

- To develop an AI-powered web application for real-time fraud detection and customer classification in e-commerce platforms, using machine learning algorithms such as Random Forest, logistic regression, and neural networks.

- To accurately identify and classify high-revenue (retained) and low-revenue (not retained) customers by analyzing transaction patterns, behavioral data, and fraud indicators, enhancing business decision-making and customer retention strategies.

# PROPOSED SYSTEM



# EXISTING METHODOLOGIES

Machine learning classifiers like Logistic Regression, Random Forest, and Neural Networks are commonly used in existing fraud detection systems. Logistic Regression offers interpretability and performs well on linearly separable data, while Random Forests provide robustness and handle nonlinearities efficiently. Neural Networks, especially deep learning models, capture complex patterns but require significant computational resources and data tuning. However, in pursuit of higher accuracy and reduced false positives.

# DRAWBACKS

**Easily Bypassed**: Advanced scammers can easily analyze these model

**Human Error**: Manual testing require human involvement so common mistakes will be caused.

**Time complexity:** Takes more time to detect the fraud.

# PROPOSED METHODOLOGIES

Gradient Boosting is an ensemble machine learning technique that builds a strong predictive model by combining multiple weak learners, typically decision trees. The idea is to sequentially train models so that each new model focuses on correcting the errors made by the previous ones. Initially, the model starts with a basic prediction (like the mean for regression or log-odds for classification). At each step, it adds a new tree trained to predict the residuals (errors) of the current model.

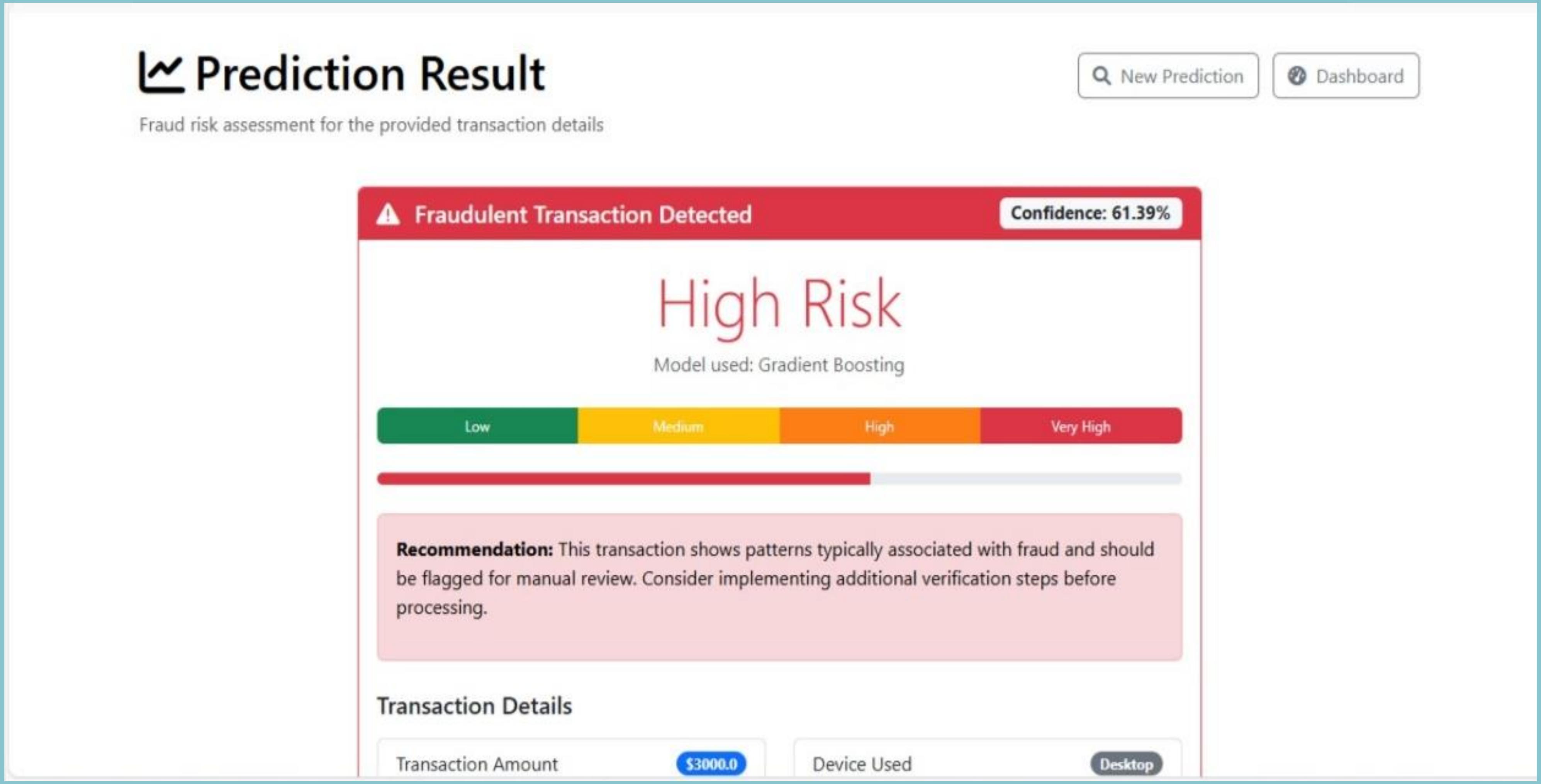
# SAMPLE CODE

```
import os
import logging
from flask import Flask, render_template, request, jsonify,
 redirect, url_for, flash
from data_processor import DataProcessor
from visualization import Visualization
from model_trainer import ModelTrainer
import pandas as pd
import numpy as np
import json
```

# DATA SET

| | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | Transactio | Customer | Transactio | Transactio | Payment N | Product Ca | Quantity | Customer | Customer | Device Us | IP Address | Shipping A | Billing Add | Is Fraudule | Account A | Transaction Hour | |
| 2 | c12e07a0- | 8ca9f102- | 42.32 | ######## | PayPal | electronic: | 1 | 40 | East Jame: | desktop | 110.87.24 | 5399 | 5399 | 0 | 282 | 23 | |
| 3 | 7d187603- | 4d158416- | 301.34 | ######## | credit card | electronic: | 3 | 35 | Kingstad | tablet | 14.73.104. | 5230 | 5230 | 0 | 223 | 0 | |
| 4 | f2c14f9d- | ccae47b8- | 340.32 | ######## | debit card | toys & gan | 5 | 29 | North Rya | desktop | 67.58.94.9 | 195 Cole | 4772 | 0 | 360 | 8 | |
| 5 | e9949bfa- | b04960c0- | 95.77 | ######## | credit card | electronic: | 5 | 45 | Kaylaville | mobile | 202.122.1 | 7609 | 7609 | 0 | 325 | 20 | |
| 6 | 7362837c- | de9d6351- | 77.45 | ######## | credit card | clothing | 5 | 42 | North Edw | desktop | 96.77.232. | 2494 | 2494 | 0 | 116 | 15 | |
| 7 | 5da506fe- | 03033baf- | 345.27 | ######## | PayPal | toys & gan | 1 | 9 | Johnsonm | desktop | 158.48.16 | PSC 3832, | PSC 3832, | 0 | 251 | 13 | |
| 8 | 47b35c5d- | 6a5305a3- | 53.69 | ######## | debit card | toys & gan | 3 | 41 | New Bren | mobile | 93.54.173. | 272 | 272 | 0 | 138 | 13 | |
| 9 | bf3db41c- | 7300dcf3- | 680.17 | ######## | debit card | electronic: | 5 | 39 | South Trac | tablet | 61.52.160. | 30470 | 30470 | 0 | 36 | 23 | |
| 10 | 8ec806af- | 25c48d47- | 126.5 | ######## | debit card | home & ga | 4 | 35 | West Henr | mobile | 6.59.118.1 | 179 Amy | 179 Amy | 0 | 188 | 4 | |
| 11 | 3b51c8cd- | 41803857- | 47.18 | ######## | bank trans | toys & gan | 4 | 19 | Taylorside | mobile | 17.185.69. | 34828 | 34828 | 0 | 245 | 22 | |
| 12 | 147217fc- | 6205d7d8- | 137.74 | ######## | PayPal | health & b | 2 | 24 | East Scott | mobile | 129.112.4 | 672 | 672 | 0 | 256 | 12 | |
| 13 | 2a2a4e1d- | 21dc4cc4- | 141.09 | ######## | PayPal | health & b | 1 | 53 | Lake Cynt | mobile | 68.178.24 | 869 | 869 | 0 | 205 | 0 | |
| 14 | c2d10730- | 1dce876f- | 95.13 | ######## | credit card | electronic: | 4 | 51 | New Eliza | mobile | 146.189.2 | 7565 | 007 | 0 | 95 | 19 | |
| 15 | 326199a2- | 3612d33f- | 260.69 | ######## | debit card | electronic: | 5 | 15 | New Aaro | tablet | 217.121.6 | 628 Cole | 0750 Jay | 0 | 158 | 2 | |
| 16 | 6dab2ff0- | c172884f- | 70.66 | ######## | debit card | home & ga | 3 | 42 | West Jam | tablet | 161.56.23 | 1437 | 1437 | 0 | 209 | 16 | |
| 17 | f77b607f- | a078ac2b- | 36.12 | ######## | PayPal | toys & gan | 2 | 39 | East Joshu | tablet | 9.65.99.20 | 85492 Hill | 85492 Hill | 0 | 301 | 14 | |
| 18 | 6760f009- | 49304d18- | 57.91 | ######## | debit card | home & ga | 3 | 33 | Annamout | mobile | 44.233.18 | 506 | 506 | 0 | 352 | 17 | |
| 19 | 15547929- | 116c1649- | 10.14 | ######## | credit card | home & ga | 1 | 24 | Howardch | mobile | 113.105.1 | 53240 | 53240 | 0 | 347 | 5 | |
| 20 | 9d84b64e- | 512c4d1d- | 261.52 | ######## | credit card | toys & gan | 1 | 37 | East Willia | mobile | 105.40.13 | 9829 | 9829 | 0 | 72 | 11 | |
| 21 | c2702a20- | c7653de4- | 258.23 | ######## | PayPal | toys & gan | 4 | 41 | Arnoldburg | tablet | 56.128.16 | 4482 Paul | 4482 Paul | 0 | 44 | 22 | |
| 22 | f675cd27- | 8bc028c7- | 438.16 | ######## | credit card | electronic: | 1 | 34 | Jenniferbe | tablet | 26.8.155.1 | PSC 8138, | PSC 8138, | 0 | 118 | 12 | |
| 23 | 9770f158- | 050cabaf- | 206.21 | ######## | debit card | electronic: | 1 | 37 | Martinezt | desktop | 138.209.1 | 25427 Hill | 25427 Hill | 0 | 60 | 19 | |
| 24 | 0dcaff26- | e35cf204- | 520.27 | ######## | PayPal | toys & gan | 2 | 18 | South Edw | mobile | 207.105.1 | 31001 | 31001 | 0 | 213 | 2 | |
| 25 | 94204afa- | 95f76bdf- | 412.96 | ######## | credit card | home & ga | 2 | 33 | Port Sean | desktop | 204.19.97. | 30888 | 30888 | 0 | 211 | 17 | |
| 26 | 220fefde- | 08b264b7- | 183.6 | ######## | bank trans | electronic: | 3 | 27 | Port Richa | tablet | 57.85.170. | 8954 | 8954 | 0 | 360 | 21 | |
| 27 | 59ac0ba0- | b11bf3de- | 112.26 | ######## | debit card | home & ga | 4 | 48 | New Dian | desktop | 138.228.1 | 043 Jean | 043 Jean | 0 | 271 | 3 | |
| 28 | 0f664ed3- | f253293d- | 23.67 | ######## | bank trans | electronic: | 5 | 35 | East Keith | desktop | 201.201.1 | 6583 | 6583 | 0 | 58 | 6 | |
| 29 | 387733c1- | f450da42- | 148.94 | ######## | PayPal | clothing | 3 | 45 | Barnesche | mobile | 79.79.204. | 907 | 907 | 0 | 297 | 0 | |
| 30 | 0ff77f0f-3 | 7eff4508- | 369.54 | ######## | debit card | toys & gan | 5 | 22 | Port Richa | tablet | 197.60.20 | 714 | 714 | 0 | 336 | 13 | |
| 31 | f9959bb7- | aeeb7e26- | 530.41 | ######## | debit card | clothing | 5 | 40 | East Willia | mobile | 104.45.24 | 8176 | 8176 | 0 | 341 | 7 | |

# RESULT ANALYSIS



# ADVANTAGES AND APPLICATION

- Scalability with High Transaction Volumes
- High Accuracy and Reduced False Positives
- Transaction Risk Scoring
- Account Takeover Detection
- Payment Fraud Detection

# CONCLUSION

The integration of Artificial Intelligence in the governance of e-commerce, AI-enhanced fraud detection serves as a strategic answer to the persistent online security challenges. Algorithms based on AI technologies, machine learning and data analytics, can identify highly sophisticated patterns of fraud in real time by efficiently sifting through enormous user behavioral data and transaction logs.

# FUTURE SCOPE

In the future, the domain of e-commerce will greatly benefit from AI technologies integrated with cybernetic security protocols for the prevention of online fraud. Upcoming innovations might include merging AI with blockchain technology for improved transparency and sharing of information between platforms without compromising security.

# REFERENCES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